

JEE MAIN REVISION MASTER: AREA UNDER THE CURVES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 170 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Area Under The Curves - Core Application	170 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Area Under The Curves - Core Application (Very High Yield — 170 PYQs)

Core Principle: Key problem pattern in Area Under The Curves.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $f:(1,\infty) \rightarrow \mathbb{R}$ be a function defined as $f(x)=(x-1/x+1)$. Let $f^{i+1}(x)=f(f^i(x)), i=1,2,\dots,25$, where $f^1(x)=f(x)$." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The area of the region $R=(x,y):xy \leq 27, 1 \leq y \leq x^2$ is equal to $\dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The area of the region $R=(x,y):xy \leq 8, 1 \leq y \leq x^2, x \geq 0$ is $\dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

Let $f:\mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x)+3f((\pi/2)-x)=\sin x, x \in \mathbb{R}$. Let the maximum value of f on $\mathbb{R}$ be α . If the area of the region bounded by the curves $g(x)=x^2$ and $h(x)=\beta x^3, \beta > 0$, is α^2 , then $30\beta^3$ is equal to ____.

Correct Answer: 16

Solution Analysis: $f(x)+3f((\pi/2)-x)=\sin x \dots (1)$ Now replace x by $(\pi/2)-x$ in (1). This is a standard NCERT step to get a second equation in the same two terms. $\Rightarrow f((\pi/2)-x)+3f(x)=\cos x \dots (2)$ Now we solve the two linear equations (1) and (2) to find $f(x)$. Multiply (1) by 3: $3f(x)+9f((\pi/2)-x)=3\sin x \dots (3)$ Subtract (2) from (3): $(3f(x)-3f(x))+(9-1)f((\pi/2)-x)=3\sin x-\cos x$ $8f((\pi/2)-x)=3\sin x-\cos x$ Now put $x \rightarrow (\pi/2)-x$ in this result to get $f(x)$: $8f(x)=3\cos x-\sin x$ $f(x)=(1/8)(3\cos x-\sin x)$ To find the maximum value,